Social Media Reach Analysis

How far content travels, who engages, and what to change: four platforms, six months

Scope	553 posts on Instagram, TikTok, Facebook, LinkedIn, January - June 2026
Records	553 clean posts (560 raw) plus monthly follower counts
Tools	Python (pandas, scipy, matplotlib), Excel (pivot tables)
Data	Synthetic, generated for learning purposes

Headline results

Total reach / engagements	4,345,906 / 297,215
Median engagement rate (by reach)	5.3%
Largest reach share	Instagram (48% of total reach)
Fastest follower growth	TikTok (+42% in 6 months)
Best format by reach / engagement	Video / Carousel
Top 10% of posts deliver	38% of all reach

1. Objective and Scope

The study answers five questions: how much reach and engagement each platform delivers, how fast audiences are growing, which post formats work best, when posting performs best, and what a small team should do differently next quarter.

Deliverables

- Cleaned post-level dataset, raw version, and monthly follower table (CSV)
- Platform, format, timing and hashtag analysis using fair, platform-adjusted comparisons
- Four charts, top posts, recommendations and limitations

Note on data

The data is synthetic, built with realistic features: a few viral posts, different platform behaviours and messy exports. Replace it with exports from each platform's analytics (Meta Business Suite, TikTok Analytics, LinkedIn Analytics) to get real findings.

2. Dataset Description

Column	Meaning
Post_ID / Platform / Post_Type	Identifier, network, and format (Video, Image, Carousel, Text, Link)
Posted_At	Publish date and time
Impressions / Reach	Total views / unique accounts reached
Likes, Comments, Shares, Saves	Engagement actions
Link_Clicks	Clicks on a link in the post
Followers_At_Post	Account followers when posted
Hashtags / Caption_Length	Hashtag count and caption characters

Sample rows (cleaned)

Post	Platform	Type	Posted	Reach	Likes	Shares	ER %
P0047	Instagram	Video	Jan 02 18:05	6,985	342	69	7.1
P0183	Instagram	Video	Mar 27 03:25	4,189	291	58	9.7
P0104	Instagram	Image	Jun 07 16:15	4,182	169	25	5.8
P0125	Instagram	Video	May 31 17:39	20,636	933	208	6.5
P0422	Facebook	Video	Apr 30 16:46	1,893	25	6	2.3
P0115	Instagram	Image	Jan 28 18:36	8,998	391	48	5.9
P0332	Facebook	Video	May 21 22:26	2,894	58	8	2.7
P0112	Instagram	Video	Mar 22 12:37	12,251	530	91	5.9

3. Data Cleaning

Six problems were found and fixed before any metric was calculated.

Issue	Count	Fix
Duplicate rows	7	Removed
Platform name variants (IG, FB, insta...)	40	Mapped to four standard names
Mixed date formats	20	Parsed each format explicitly
Missing impressions	10	Estimated: reach x platform median impressions-per-reach
Reach greater than impressions	10	Reach capped at impressions (impossible otherwise)
Missing shares	15	Estimated from median share rate of same platform and format

Result: 560 raw rows became 553 posts. Three of these fixes are estimates; with real exports, missing values should be re-pulled from the platform instead of imputed.

4. Metric Definitions and Method

Metric	Formula	Why
Engagement	Likes + comments + shares + saves	Total actions
Engagement rate (ER)	Engagement / reach	Quality of response among people who saw it
Reach rate	Reach / followers at posting	How far a post travels beyond or within the audience
Frequency	Impressions / reach	Average views per person reached
CTR	Link clicks / impressions	Traffic driven
Share and save rate	Shares or saves / reach	Strong interest signals

Fair comparisons. Platforms behave differently: TikTok reach can exceed followers while Facebook reach is a small share. Comparing raw numbers across platforms would only rank the platforms. So format and timing results use **indexes**: reach and engagement rate are divided by the median for the same platform (format comparisons) or the same platform and format (timing comparisons). A value of 1.0 means typical, 1.2 means 20% better than typical.

Medians, not means. A few viral posts distort averages: the mean reach is 1.5x the median. Medians describe a typical post, and the Kruskal-Wallis test (which does not assume a bell-shaped distribution) is used for group comparisons.

5. Analysis

5.1 Platform performance

Platform	Posts	Total reach	Share	Median reach	Median ER %	Median reach rate %
Instagram	190	2,098,641	48%	7,700	6.2	57
TikTok	132	1,883,716	43%	11,804	7.0	166
Facebook	121	161,954	4%	947	2.6	11
LinkedIn	110	201,595	5%	1,150	4.0	27

Instagram delivers the largest share of reach (48%), while TikTok has the highest median engagement rate (7.0%). Reach rate above 100% on a platform means a typical post reaches more accounts than the account has followers.

5.2 Audience growth

Platform	Jan 1	Jun 30	Net gain	Growth	Avg monthly
Instagram	12,000	13,562	1,562	+13%	+2.1%
TikTok	5,000	7,091	2,091	+42%	+6.0%
Facebook	8,000	8,280	280	+3%	+0.6%
LinkedIn	3,500	4,184	684	+20%	+3.0%

TikTok grew fastest at 6.0% a month; Facebook was slowest at 0.6%. Growth shown is of the account, which does not show whether it was caused by content or by other activity.

5.3 Post format

Median index by format across platforms (formats with at least 15 posts):

Format	Posts	Reach index	ER index	Share rate %	Save rate %
Video	249	1.17	1.00	0.91	0.46
Carousel	74	1.05	1.34	0.74	1.04
Text	55	0.98	0.97	0.35	0.30
Image	119	0.86	0.98	0.47	0.34
Link	56	0.54	0.78	0.32	0.17

Formats differ significantly in reach ($p < 0.001$) and in engagement rate ($p < 0.001$) (Kruskal-Wallis). Video posts reach the most people; Carousel posts earn the highest engagement rate. Reach and engagement rate are different strengths, so the best format depends on whether the goal is awareness or response.

Median engagement rate (%) by format and platform, with number of posts:

Format	Instagram	TikTok	Facebook	LinkedIn
Carousel	8.3 (n=53)	-	-	5.8 (n=21)
Image	5.1 (n=61)	-	3.2 (n=38)	4.3 (n=20)
Link	-	-	2.0 (n=31)	3.0 (n=25)
Text	-	-	2.5 (n=25)	4.1 (n=30)
Video	6.1 (n=76)	7.0 (n=132)	2.8 (n=27)	3.6 (n=14)

5.4 Posting time

Each post's engagement rate is divided by the median for its own platform and format, then grouped by time slot. The heat map in section 7 shows weekday by time of day. Across all posts, time of day shows a significant effect ($p < 0.001$, Kruskal-Wallis). Best and worst slots by platform (slots with at least 12 posts):

Platform	Best slot	Index	Posts	Worst slot	Index
Instagram	Evening (17-21)	1.18	56	Early morning (5-9)	0.78
TikTok	Evening (17-21)	1.19	38	Late morning (9-12)	0.93
Facebook	Afternoon (12-17)	1.06	50	Early morning (5-9)	0.81
LinkedIn	Late morning (9-12)	1.16	26	Night (21-5)	0.89

Weekday vs weekend (median timing index): Instagram 1.00 vs 1.00; TikTok 1.03 vs 0.98; Facebook 1.00 vs 1.00; LinkedIn 1.08 vs 0.82.

5.5 Hashtags

Hashtags per post	Posts	Reach index	ER index
0	37	0.92	0.95
1-3	215	0.93	1.02
4-6	166	1.07	0.99
7-10	118	1.06	1.02
11+	17	0.92	1.02

Reach is highest at 4-6 hashtags per post (index 1.07). Hashtag count is also tied to platform, because Instagram posts use far more than Facebook, but the index removes the platform-level difference.

5.6 Other relationships (Spearman correlation across posts)

Pair	rho	Significance
Hashtag count vs reach index	+0.12	p = 0.005
Caption length vs engagement-rate index	-0.05	p = 0.207
Reach index vs engagement-rate index	+0.19	p < 0.001
Shares per reach vs reach index	+0.13	p = 0.002

5.7 Concentration of reach

The top 10% of posts account for **38%** of all reach and the top 1% for 10%. Results depend heavily on a handful of posts, so understanding why those few worked matters more than polishing the average post.

6. Top Posts

Top 10 by reach

Post	Platform	Type	Reach	Eng.	ER %	Shares
P0172	Instagram	Video	235,901	9,662	4.1	1,358
P0257	TikTok	Video	47,096	3,135	6.7	436
P0176	Instagram	Video	43,020	4,724	11.0	690
P0316	TikTok	Video	41,627	6,760	16.2	982
P0269	TikTok	Video	39,914	6,341	15.9	934
P0116	Instagram	Video	38,574	4,416	11.4	651
P0271	TikTok	Video	38,553	3,070	8.0	423
P0271	TikTok	Video	38,553	3,070	8.0	423
P0243	TikTok	Video	37,315	2,276	6.1	325
P0217	TikTok	Video	35,649	2,370	6.6	328

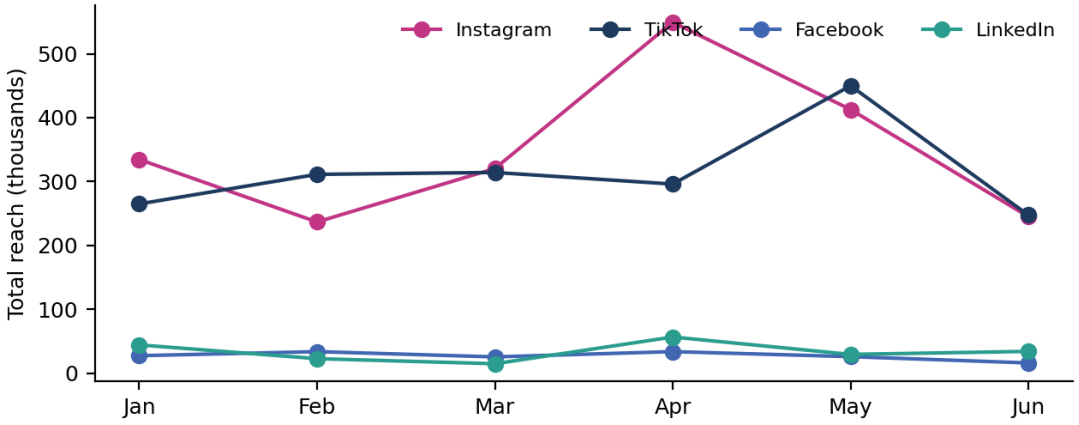
Top 10 by engagement rate (posts with at least 500 reach)

Post	Platform	Type	Reach	Eng.	ER %	Saves
P0075	Instagram	Carousel	5,199	909	17.5	111
P0316	TikTok	Video	41,627	6,760	16.2	495
P0269	TikTok	Video	39,914	6,341	15.9	437
P0033	Instagram	Carousel	6,978	941	13.5	132
P0060	Instagram	Carousel	3,529	468	13.3	51
P0036	Instagram	Carousel	5,056	670	13.3	79
P0255	TikTok	Video	18,599	2,461	13.2	145
P0220	TikTok	Video	11,097	1,437	12.9	110
P0107	Instagram	Carousel	18,765	2,422	12.9	335
P0228	TikTok	Video	19,794	2,536	12.8	193

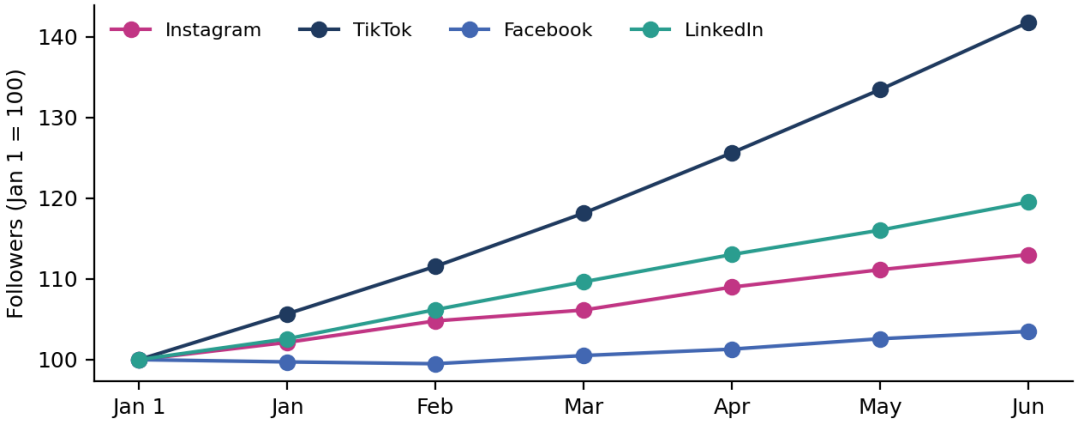
The top-reach list is led by TikTok posts (7 of 10) and Video content (10 of 10).

7. Visualizations

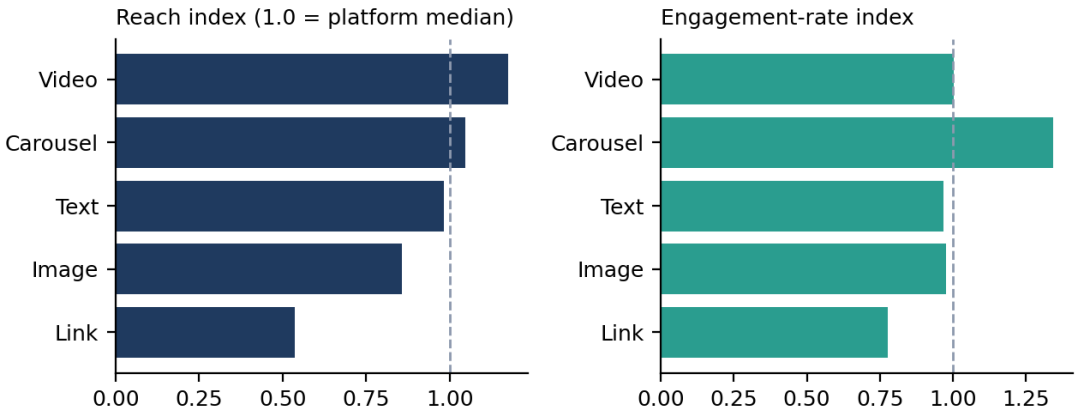
Monthly Reach by Platform

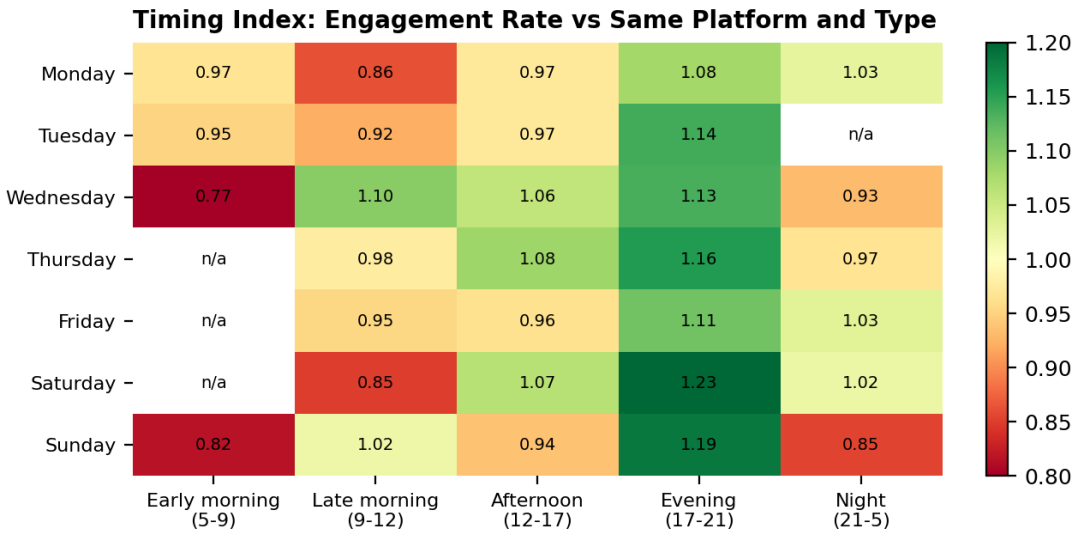


Follower Growth, Indexed



Post Type Performance (platform-adjusted)




Timing Index: Engagement Rate vs Same Platform and Type
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
0.97 0.86 0.97 1.08 1.03
0.95 0.92 0.97 1.14 n/a
0.77 1.10 1.06 1.13 0.93
n/a 0.98 1.08 1.16 0.97
n/a 0.95 0.96 1.11 1.03
n/a 0.85 1.07 1.23 1.02
0.82 1.02 0.94 1.19 0.85
Early morning (5-9)
Late morning (9-12)
Afternoon (12-17)
Evening (17-21)
Night (21-5)
1.20
1.15
1.10
1.05
1.00
0.95
0.90
0.85
0.80

8. Recommendations

1. Match format to the goal. For reach, lean on Video (reach index 1.17); for engagement, Carousel (ER index 1.34). Link posts have the lowest reach index (0.54), so use them sparingly.

2. Post in each platform's best slot. Instagram: evening (17-21). TikTok: evening (17-21). Facebook: afternoon (12-17). LinkedIn: late morning (9-12). Test these for a month, since time of day explains only part of the variation.

3. Invest where audience growth is fastest. TikTok grew +42% in six months, but check whether reach and engagement keep pace as the audience gets larger.

4. Use hashtags in moderation. The reach index is highest at 4-6 hashtags per post (1.07) and similar at 7-10 (1.06); posts with 11 or more score 0.92, though only 17 posts fall in that group, so treat it as a hint.

5. Study the outliers. The top 10% of posts deliver 38% of reach. Review the top ten posts for hooks, topics and formats, and repeat what they share.

Metrics to track next quarter: median reach rate, median engagement rate, share and save rate, monthly follower growth, and the share of posts in the best time slot.

9. Limitations

- Synthetic data; real accounts have paid promotion, collaborations, algorithm changes and events that confound results.
- Six months and about 100 to 200 posts per platform: small cells in the timing heat map are noisy.
- Impressions, reach and followers are defined differently on each platform, so cross-platform numbers are only roughly comparable.
- Associations (format, timing, hashtags) are not experiments; format and time were not randomly assigned.
- Follower growth cannot be attributed to individual posts with this data.

Appendix: Python script (core analysis)

```
import pandas as pd
from scipy import stats
d = pd.read_csv("social_posts_clean.csv", parse_dates=["Posted_At"])
d["Engagement"] = d.Likes + d.Comments + d.Shares + d.Saves
d["ER"]        = d.Engagement / d.Reach * 100
d["ReachRate"] = d.Reach / d.Followers_At_Post * 100

# platform-adjusted indexes (1.0 = typical for that platform)
d["ReachIdx"] = d.Reach / d.groupby("Platform").Reach.transform("median")
d["ERIdx"]    = d.ER / d.groupby("Platform").ER.transform("median")
d["TimeIdx"]  = d.ER / d.groupby(["Platform","Post_Type"]).ER.transform("median")

by_type = d.groupby("Post_Type")[["ReachIdx","ERIdx"]].median()
groups  = [g.ERIdx.values for _, g in d.groupby("Post_Type")]
print(stats.kruskal(*groups))

def bucket(h):
    return ("Early morning" if 5 <= h < 9 else "Late morning" if h < 12 else
            "Afternoon" if h < 17 else "Evening" if h < 21 else "Night")
d["Bucket"]  = d.Posted_At.dt.hour.map(bucket)
d["Weekday"] = d.Posted_At.dt.day_name()
heat = d.pivot_table(index="Weekday", columns="Bucket",
                     values="TimeIdx", aggfunc="median")

top10_share = d.Reach.nlargest(int(len(d) * .1)).sum() / d.Reach.sum()
```

Excel equivalents

- Engagement rate: =(Likes+Comments+Shares+Saves)/Reach
- Median by group: PivotTable cannot take medians; use =MEDIAN(IF(PlatformRange="Instagram",ERRange)) as an array formula, or Power Pivot
- Time of day: =HOUR(Posted_At); weekday: =TEXT(Posted_At,"dddd")
- Heat map: PivotTable (Weekday x Time slot) with Conditional Formatting > Color Scales